

Dynamics Test

Ethan Wang

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Blocks

1. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 144N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:
 - (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block
2. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 224N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:
 - (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block
3. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 30N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:
 - (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block
4. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 12N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:
 - (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block
5. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 234N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:
 - (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block
6. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 423N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:
 - (a) The acceleration of the system
 - (b) The acceleration of each block

- (c) The net force on each block
 - (d) The contact forces for each block
7. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 116N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:
- (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block
8. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 126N force acts on block 2 from the right, and the surface can be considered frictionless. Calculate:
- (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block
9. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 306N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:
- (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block
10. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 450N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:
- (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block